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Study for Integrating IoT-IDS Datasets: Machine and Deep Learning for Secure IoT Network System

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ABSTRACT

The rapid expansion of Internet of Things (IoT) devices has introduced a new phase of inter connectivity and convenience, while also presenting notable security obstacles. This research paper investigates novel methodologies for enhancing the security of IoT networks by using machine learning and deep learning methodologies. In order to accomplish this objective, two well acknowledged datasets, namely UNSW-NB15 and BoT-IoT, are used for the purpose of constructing and assessing resilient security models. The UNSW-NB15 dataset is well recognized for its extensive compilation of network traffic data, while the BoT-IoT dataset is especially designed to facilitate study on security issues related to the IoT. These datasets provide a wide array of attack scenarios and network traffic patterns, therefore enhancing the diversity of available resources for researchers in this field. By using these datasets, our research endeavours to examine many crucial facets of security inside IoT networks. Initially, machine learning techniques are used for the purpose of conducting intrusion detection on IoT networks. This stage encompasses the categorization of network traffic into two distinct categories: normal and malicious. This process facilitates the prompt detection and reaction to potential threats in real-time. Deep learning methodologies demonstrate exceptional proficiency in capturing complicated patterns and behaviours inherent in IoT network traffic, hence enhancing the capacity to detect intricate and dynamic security risks. In order to assure the practical usability of the presented models, we take into account variables such as computing efficiency and scalability. This is particularly important since IoT devices often have limited resources available. Furthermore, we investigate approaches aimed at enhancing the interpretability of models, therefore offering valuable insights into the underlying decision-making mechanisms, ultimately fostering trust and transparency.

This study makes a valuable contribution to the emerging area of IoT security by showcasing the effectiveness of machine learning and deep learning techniques in enhancing the protection of IoT networks. Our work contributes to the advancement of knowledge about IoT risks and the construction of adaptive, resilient security mechanisms by using the extensive UNSW-NB15 and BoT-IoT datasets.

KEYWORDS

Intrusion, ML, DL, security, SVM, CNN

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1 INTRODUCTION

Emergence of the IoT has initiated a period characterized by unparalleled interconnectedness and automation, fundamentally transforming our interactions with technology and the surrounding environment [1]. The IoT encompasses a wide range of devices, including smart household appliances and industrial sensors, which have been seamlessly integrated into our everyday routines. This integration has resulted in notable improvements in terms of convenience, efficiency, and productivity [2]. The extensive integration of IoT technology has, though, brought to light weaknesses in network security, giving rise to significant apprehensions over the safeguarding of confidential information and the reliability of linked systems [3]. The issue of security in IoT networks is of utmost importance due to the increased number of devices, which in turn expands the potential for attacks. This makes these networks attractive targets for those with bad intent [4]. Sophisticated measures are necessary in order to reduce the security threats associated with IoT ecosystems and ensure the preservation of their integrity [5]. This study aims to tackle the aforementioned difficulties by using machine learning and deep learning methodologies, which will be applied to two separate but complimentary datasets: the UNSW-NB15 dataset [6] and the BoT-IoT dataset [7]. IoT networks encounter a wide range of security concerns, including several aspects such as: IoT networks are vulnerable to a range of attack vectors, including the transmission of malware, distributed denial-of-service (DDoS) assaults, impersonation of devices, and breaches of data security [8]. One of the primary challenges in implementing solid security measures for IoT devices is the presence of resource constraints, which sometimes restrict their processing capabilities [9].

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The presence of diverse devices, protocols, and communication standards within IoT ecosystems gives rise to heterogeneous and intricate security demands. The protection of user information is crucial when it comes to IoT devices, since they often handle sensitive data. Therefore, it is essential to include strong privacy measures to ensure data privacy [10]. These techniques has the capability to evaluate large quantities of data in real-time, hence facilitating the identification of abnormalities and malicious behaviours that might otherwise prove difficult to discover with conventional rule-based approaches [11]. The use of conventional ML models, such as DT, SVM, NB and RF, is prevalent in the domain of intrusion detection and classification tasks. These models has the capability to detect well-known assaults by analyzing patterns and extracting information from network traffic data. Deep learning, a branch of machine learning, has shown remarkable proficiency in capturing intricate and non-linear associations present in datasets. This is primarily achieved via the use of advanced models such as CNNs. These models are especially well-suited for the identification of unknown and emerging dangers in IoT network traffic. The availability of appropriate datasets is crucial for doing research [12-13].

The UNSW-NB15 dataset is well recognized as a prominent benchmark in the field of network security and comprises a range of data related to network traffic, including legitimate along with malicious actions. The BoT-IoT platform offers a comprehensive assortment of network traffic situations and attack patterns, facilitating researchers in their exploration and development of solutions to safeguard IoT settings from emerging threats. The present study provides a significant addition to ongoing efforts to improve the reliability and safety of IoT ecosystems. It does this by developing and evaluating models that can effectively and rapidly identify both existing and potential threats. The quest of securing IoT technology is crucial as it continues to transform our linked world. This study represents a significant advance in that direction.

2 LITERATURE REVIEW

A lightweight CNN was used to identify different forms of spoofing assaults. The dataset used for this objective was ASVspoof17, to decrease the features related to input and also for the optimization of the speed for processing. The primary objective of the research was to enhance precision, while also recognizing the possible limitation of reduced system performance [14]. In their study, the researchers used a deep learning classifier and utilized the AWID dataset to detect several types of threats, including Flooding, Injection, and Impersonation, inside the framework of 5G-IoT. The accuracy rate of the classifier was found to be 99.8% by the use of an autoencoder deep neural network [15]. The datasets in question use the Multilayer Perceptron (MLP) methodology to evaluate the efficacy of several ML algorithms [16], The assessment measures used in this study were Accuracy and F1-score. The tests were done by the authors using the CICIDS2017 dataset. In a similar vein, the scholars in reference [17-18] used distributed DL methodologies to detect instances of cyber assaults inside IoT networks. The researchers used the NBaIOT dataset and implemented a CNN to evaluate the accuracy and F1 score metrics in the detection of phishing and bot assaults inside IoT networks. The CNN methodology is used for the purpose of identifying intrusion attacks inside IoT networks, with a particular focus on detecting DoS attacks. The evaluation of the accuracy parameter is performed using the BoT-IoT dataset, with the TensorFlow tool being used for this task. Moreover, the primary aim is to use this methodology in real-time to effectively counteract assaults [19]. The objective of this research is to examine the efficacy of feature selection strategies, within the domain of intrusion detection. The chosen characteristics are then used in a deep learning methodology, especially a Convolutional Neural Network (CNN), to identify several forms of assaults using the KDD dataset. The main objective of this investigation is to assess the efficacy of detecting these assaults. The decision was taken to use intelligent agents with the aim of enhancing the efficiency of data transmission [20]. The use of a CNN inside IoT networks for the purpose of anomaly detection is a well recognized and prevalent methodology. The use of this deep learning-based model has been seen in many datasets. The aforementioned investigations used CNN1D, CNN2D, and CNN3D architectures, which yielded notable accuracy rates of around 99.74%, 99.42%, and 99.03% correspondingly [21]. Similarly, Ge et al. [22] introduced a technique for intrusion detection that employs a Bi-LSTM RNN. The primary aim of their research was to effectively categorize patterns into two distinct groups: normal instances and attack instances. The implanted model has undergone a process of training. This study proposes a novel methodology for creating connection inside IoT networks, using the principles of DL. The user's content exhibits a lack of detail and clarity. The model exhibited a notable degree of accuracy when used to binary classification tasks; however, its performance was not as precise when employed in multiclass classification tasks. In the same way, Pecori et al. [23] did a research whereby they curated a dataset including both legitimate and fraudulent network traces specific to the IoT by amalgamating several datasets. The dataset has four unique categories of attack, in addition to a normal category. To enhance performance, the researchers used a neural network design including seven hidden layers and performed training for a cumulative duration of 50 epochs. Notwithstanding these efforts, the achieved levels of accuracy, recall, and F1 score were considered inadequate when used in both binary and multiclass classification endeavours. In the study, Idrissi et al. [24] conducted an inquiry with the aim of discovering vulnerabilities and security risks inside the domain of the IoT. In the lieu of, Li et al. [25] provide an innovative model framework for the selection of characteristics and detection of abnormalities in smart cities within the context of the IoT. The suggested model aims to increase the accuracy and efficacy of the intrusion detection process by integrating deep machine learning with intrusion detection technologies. The researchers provide a methodology for assessing the effectiveness of migrating learning models and determining the optimal system characteristics. The use of the UNSW-NB15 dataset has yielded a detection accuracy of 95% in the domain of identifying attacks in the Internet IoT. The researchers presented a theoretical framework for the identification of Botnets in their study. The used methodology included the utilization of a CNN using a one-class classification technique. The model included features derived from the system call graph. The model's accuracy was assessed to be 97%, while the F-measure was found to be 98.33%. The user provided a numerical reference [26].

3 ML AND DL CLASSIFIERS

3.1 SVM

SVM have garnered considerable attention since its inception in the 1960s by Vladimir Vapnik and his collaborators. This may be attributed to their efficacy in dealing with linear and non-linear data, their resilience in high-dimensional domains, and their capability to identify a distinct separation boundary between different classes [27]. Consequently, the transformed data may be separated by a linear hyperplane. Polynomial, Radial Basis Function (RBF), and Sigmoid kernels are often encountered in many computational applications. The concept of support vectors is a fundamental component in the field of machine learning and statistical classification [28]. Also, Support vectors refer to the specific data points within a dataset that are positioned in close proximity to the decision boundary, also known as the hyperplane. The location and direction of the hyperplane are determined by these crucial points [29].

3.2 CNN

CNNs belong to a category of deep learning models that draw inspiration from the visual processing skills shown by the human brain. Emerged as a fundamental component in the field of computer vision, serving as a crucial tool for a wide range of tasks. In addition to their prominent role in image analysis, CNNs have also shown their use in other fields like natural language processing and voice identification [30]. The central point or essential element is that CNN use convolutional layers to perform a scanning operation on input data by using filters, also known as kernels, that are capable of being adjusted via the learning process. These filters are capable of detecting various patterns and characteristics within the input, making them very efficient for jobs related to image processing. Pooling layers are a fundamental component of CNNs used for image recognition tasks. There exists several layers used by CNN. Pooling layers, such as max-pooling, serve the purpose of decreasing the spatial dimensions of the data while preserving crucial information. This approach contributes to the reduction of computing complexity and the mitigation of overfitting [31]. Hierarchical Feature Learning is a computational approach that involves the extraction and representation of features in a hierarchical manner. CNNs has the ability to autonomously acquire hierarchical representations of input. The lower layers of the neural network are responsible for capturing basic elements such as edges, whilst the upper layers integrate these information to identify more intricate patterns. CNNs demonstrate exceptional performance in tasks that include structured input, namely in domains such as image and audio processing. One advantage of using machine learning algorithms is their ability to autonomously acquire significant features from the provided data, hence eliminating the need for manual feature engineering. CNN designs, such as VGG, ResNet, and Inception, have undergone pre-training on extensive datasets and may afterwards be fine-tuned to cater to particular applications. There are several drawbacks associated with this phenomenon. The process of acquiring knowledge, skills, and competencies through systematic instruction and practice. CNNs sometimes impose a significant computational burden, necessitating the use of high-performance hardware or cloud-based computing resources. CNNs may need extensive labelled datasets in order to achieve optimal training outcomes. The complexity of learnt representations poses a challenge in terms of the interpretability of deep CNNs [32].

4 STEPS FOR PROPOSED METHODOLOGY

The following measures have been implemented in order to attain our objective:

- **Collection and Pre-processing:**The first phase of the research involves the collection and pre-processing of data. To ensure the validity and dependability of our research, we procured a representative sample of normal and attack network traffic from the overall population of two pre-established datasets. It is important to accurately assign the appropriate attack classifications to the information provided. Prior to processing, it is recommended to carry out pre-processing procedures such as cleaning, filtering, and normalization operations on the data [33]. The process of extracting significant attributes from network traffic data is a fundamental aspect of feature engineering. The creation of a deep learning model may include several relevant aspects such as statistical measures, traffic patterns, and packet size distributions [34]. In addition, it is possible to use feature selection and dimensionality reduction approaches.
- **Selection of the Model:** To effectively address the classification difficulty, it is essential to meticulously choose a suitable model. In addition, it is important to consider the architectural aspects of the network, including the number of layers, the number of units per layer, the activation functions used, and any regularization techniques included, such as dropout and batch normalization.
- **Model Selection and Architecture:**It is recommended to partition the dataset into distinct collections, namely the training, validation, and testing sets, as a crucial step in the training procedure. The use of data augmentation techniques is a very effective approach for enhancing the variability of the training dataset. In our proposed model, an appropriate loss function will be selected, such as cross-entropy, and an optimization method will be used, such as Adam, RMSprop, or other comparable optimizers. In order to mitigate the risks associated with overfitting and underfitting, it is essential to consistently evaluate the effectiveness of the validation set.
- **Model Training:**The training process involves training model on the designated dataset, using the predetermined parameters. During the evaluation phase, it is essential to assess the performance of the trained model on the testing set using suitable metrics, including accuracy, precision, recall, and F1-score. The use of a confusion matrix is employed to assess the efficacy of a model in detecting different forms of attacks, hence unveiling its capabilities and limitations in this regard.
- **Post-processing and deployment:** For the prediction process it is recommended to use various post-processing techniques, such as threshold modification or ensembling. Once the model has generated satisfactory outcomes, it is essential to deploy it in a real-world context and consistently monitor its performance in a production environment [35]. The ongoing process of enhancing and retraining the model in order

to adapt to changing threat kinds and network dynamics is often referred to as continuous improvement. It is important to consistently monitor the performance of the model in its respective area and gather supplementary labelled data in order to guarantee its sustained effectiveness.

- **Documentation:** The process of documentation and reporting involves systematically recording each step of the used technique, including data preparation, model architecture, hyperparameters, and assessment results. The aforementioned documentation is seen as advantageous in facilitating the process of replication, fostering cooperation among individuals, and facilitating future advancements. It is of utmost importance to bear in mind that the efficacy of a deep learning model hinges upon a multitude of factors, including domain expertise, data integrity, model architecture, and iterative refinement, as shown in Figure 1.

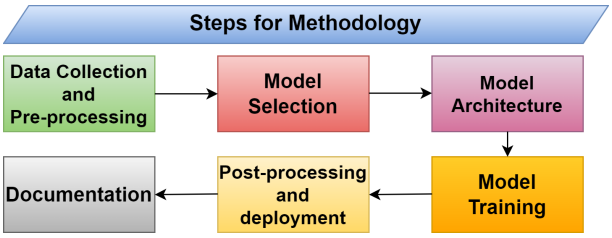

Figure 1: Steps for Proposed Methodology

5 PROPOSED METHODOLOGY

Three preprocessing procedures, namely data imputation or cleaning, discretization, and normalization, have been used. The initial step in data pre-processing is the merging of two datasets, using the 'label' column. Phrase "integrated dataset" denotes a collection of observations that arises from the amalgamation of different datasets. In the current combination of dataset, there are instances when some variables absence of values. The data imputation is a crucial step in addressing the existence of incomplete and redundant information. In order to provide accurate and reliable results, the handling of data that is vacant might include several methods, such as retaining them as empty or manually imputing values.

To apply discretization to the combined merged set of data, the use of cut points to partition an array of occurrences of a quantifiable attribute into a limited amount of ranges that demonstrate substantial coherence of class.Discretization of data refers to the process of replacing numeric values of the attributes with that of the intervals. The concept of interdependency among attributes of class pertains to the correlation between a class and its corresponding attributes within a certain context.

The range of highest values exhibits significant variability. To enable efficient arithmetic processing and dimensionality reduction, a standardized processing approach is used, whereby the value range of each feature is evenly and linearly mapped into the interval [0,1]. Once the dataset has been populated with numerical values, consisting of attributes and labels, the minimum value of the attribute can be determined using min(a), while the maximum value can be obtained using max(a). The disparity between these maximum and minimum values is significant, making it suitable for the application of the min-max normalization technique.

After pre-processing of data, the data is splitted by 80-20% rule. The training set is 80% and testing set is 20%.

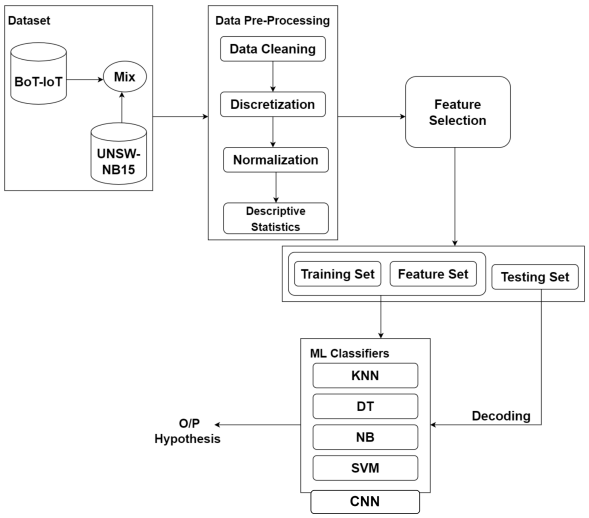

Figure 2: Proposed Methodology

6 RESULTS AND DISCUSSION

The following experiment has been done on oem-PowerEdge-T-20 device, dell inc. PowerEdge T20, Intel Xeon(R) CPU-E3-1225v3 @ 3.20Ghzx4 processor. Also, this section will discuss the results of all five classifiers such as KNN, DT, NB, SVM and CNN on the basis of binary and multiclass classification. Each one is discussed below.

6.1 Binary classification for detection of attacks

Initially, a series of tests were done with the aim of accurately categorizing assault incidents within the framework of binary categorization. In a same manner, Table 2 illustrates that in the first endeavour to construct a binary classification model, KNN outperforms DT and NB, achieving an accuracy rate of 97.2%. The reason for the degradation in the performance of DT is due to their ability to categorize both category Just like in the case of NB, while comparing KNN, it was seen that the performance of KNN over DT improved by 3.03% as a result of the determination that the independence assumptions of the characteristics were insufficient. The SVM algorithm has been assessed for its efficacy in handling non-linear data, and the findings indicate that it surpasses the KNN algorithm by a margin of 2.03%. This classification process has led to an overall relative enhancement of 5.19 %. Also CNN is showing the highest accuracy of 99.99% in detecting whether the combined dataset is having normal class or malicious class attacks. Furthermore, no irregularities have been seen while examining the binary labels.

Table 1: Binary classification for detection of attacks

ML/DL Classifier	*Accuracy*
CNN	99.99%
SVM	99.24%
NB	95.02%
DT	94.34%
kNN	97.2%

6.2 Multiclass classification for detection of attacks

Multi-class classification issues are often considered to be more intricate compared to binary classification difficulties, and attaining enhanced outcomes is a more challenging endeavour. Table 3 presents empirical evidence showcasing the enhanced effectiveness of the KNN algorithm, achieving an accuracy rate of 85.60% when using a value of k=2. KNN is a well-suited classification method for datasets including several categories, When the radius of the target data points is enlarged, it becomes challenging to classify the non-linear dataset. The maximum depth of decision trees exhibited considerable variation in relation to the amount of the training samples. DT algorithms are capable of performing classification tasks efficiently, without the need for extensive computational resources. Additionally, they provide a valuable visual representation of the importance of different variables in identifying the target class within the dataset. In order to mitigate the issue of overfitting, a predetermined threshold of data points is used to construct a terminal node. Additionally, the restriction of the tree to a certain maximum number of terminal nodes serves as a means of regulating the length of the route from the root to the leaf. The experimental results indicate that when max term nodes=2 has been used. NB algorithm has superior performance when applied to categorical input variables as opposed to numerical variables. However, it does not exhibit optimal performance when confronted with complex models that lack distinct and well-separated categories. Without the use of an appropriate refining approach, issues with a probability of zero have the potential to provide inaccurate outcomes. SVMs have exceptional generalization capabilities and provide a significant reduction in calculation time. The hyperparameter C, which is sometimes referred to as the cost function or error penalty parameter, is used for the purpose of error and misclassification management. From a technical standpoint, it should be noted that the parameter referred to as "Gamma" in the context of SVM is really associated with the kernel's perimeter, rather than being a direct SVM parameter. The RBF kernel alters the degree of similarity required between two locations. Smaller values of the hyperparameter C lead to less computational time and a reduced penalty for misclassified points, hence yielding a wider margin-based decision boundary. Conversely, greater values of C tend to produce an overfitted model. Elevated gamma values impose limitations on the radius of effect, confining it only to support vectors. As the value of gamma grows, the accuracy of the model tends to deteriorate. The best values of the parameters C and gamma have been discovered after a comprehensive and thorough analysis. At the parameter values of C=0.1 and gamma=0.1, the SVM has a classification accuracy of 91.28 percent. Also CNN is showing the 97.24% accuracte results in detecting the types of multiple classes of attacks. Figure 3 is showing the binary and multiclass classification of attack detection on both the datasets.

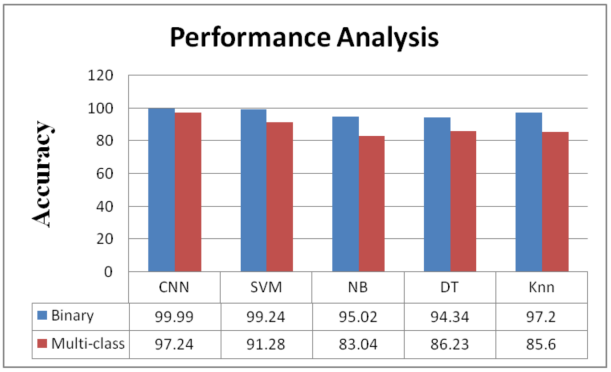

Figure 3: Binary and Multiclass Classification of Attack Detection

Table 2: Multiclass classification for detection of attacks

ML/DL Classifier	*Parameters*	*Accuracy*
CNN	-	97.24%
SVM	C=0.1 and γ=0.1	91.28%
NB	log _probability zero observation	83.04%
DT	max _term _nodes=2	86.23%
kNN	k=2	85.60%

The potential biases and limitations in the dataset that could impact the generalizability of the model are:

- The dataset focuses on a specific set of attacks such that there are no chances to generalize the novelty or evolving threats.
- Labels assigned to instances in the dataset should be unambiguous. Ambiguous or subjective labeling may introduce noise and impact the model's ability to accurately classify attacks.

7 CONCLUSION

The amalgamation of ML and DL, in conjunction with datasets such as UNSW-NB15 and BoT-IoT, facilitates the development of adaptable and robust security protocols. By acknowledging and rectifying the significant security vulnerabilities and capitalizing on the advantages of these sophisticated methodologies, effectively seize the potential benefits offered by the IoT while ensuring the protection of confidentiality, integrity, and availability of networked systems. Among five classifiers which have been analyzed for providing security CNN gave the best results in detection of attacks for both binary and multi class classification. For future study real time detection, multi label classification and specific detection of an attack are the fields to work upon.

REFERENCES

[1] Farooq, M.U., Waseem, M., Mazhar, S., Khairi, A. and Kamal, T., 2015. A review on internet of things (IoT). International journal of computer applications, 113(1), pp.1-7.

[2] Kranenburg,V. R. and Bassi, A., 2012. IoT challenges. Communications in Mobile Computing, 1(1), p.9.

[3] Zhao, J.C., Zhang, J.F., Feng, Y. and Guo, J.X., 2010, July. The study and application of the IOT technology in agriculture. IEEE, In 2010 3rd international conference on computer science and information technology (Vol. 2, pp. 462-465).

[4] Yang, Z., Yue, Y., Yang, Y., Peng, Y., Wang, X. and Liu, W., 2011, July.IEEE, Study and application on the architecture and key technologies for IOT. In 2011 International Conference on Multimedia Technology (pp. 747-751).

[5] Liu, T. and Lu, D., 2012, August. The application and development of IoT.IEEE, In 2012 International symposium on information technologies in medicine and education (Vol. 2, pp. 991-994).

[6] Ahmad, M., Riaz, Q., Zeeshan, M., Tahir, H., Haider, S.A. and Khan, M.S., 2021. Intrusion detection in internet of things using supervised machine learning based on application and transport layer features using UNSW-NB15 data-set. EURASIP Journal on Wireless Communications and Networking, 2021(1), pp.1-23.

[7] Peterson, J.M., Leevy, J.L. and Khoshgoftaar, T.M., 2021, August. A review and analysis of the bot-iot dataset. In 2021 IEEE International Conference on Service-Oriented System Engineering (SOSE) (pp. 20-27). IEEE.

[8] Hassan, W.H., 2019. Current research on Internet of Things (IoT) security: A survey. Computer networks, 148, pp.283-294.

[9] Sharma, N., Shamkuwar, M. and Singh, I., 2019. The history, present and future with IoT. Internet of things and big data analytics for smart generation, pp.27-51.

[10] Zhang, Z.K., Cho, M.C.Y., Wang, C.W., Hsu, C.W., Chen, C.K. and Shieh, S., 2014, November. IoT security: ongoing challenges and research opportunities. In 2014 IEEE 7th international conference on service-oriented computing and applications (pp. 230-234).

[11] Gubbi, J., Buyya, R., Marusic, S. and Palaniswami, M., 2013. Internet of Things (IoT): A vision, architectural elements, and future directions. Future generation computer systems, 29(7), pp.1645-1660.

[12] Lee, I. and Lee, K., 2015. The Internet of Things (IoT): Applications, investments, and challenges for enterprises. Business horizons, 58(4), pp.431-440.

[13] Saharkhizan, M., Azmoodeh, A., Dehghantanha, A., Choo, K.K.R. and Parizi, R.M., 2020. An ensemble of deep recurrent neural networks for detecting IoT cyber attacks using network traffic. IEEE Internet of Things Journal, 7(9), pp.8852-8859.

[14] Vinayakumar, R., Alazab, M., Soman, K.P., Poornachandran, P., Al-Nemrat, A. and Venkatraman, S., 2019. Deep learning approach for intelligent intrusion detection system. IEEE Access, 7, pp.41525-41550.

[15] Duraibi, S., 2021. A Secure Lightweight Voice Authentication System for IoT Smart Device Users (Doctoral dissertation, University of Idaho).

[16] Borgesen, M.E. and Kholidy, H.A., 2020. Evaluating Variant Deep Learning and Machine Learning Approaches for the Detection of Cyberattacks on the Next Generation 5G Systems.

[17] Koroniotis, N., 2020. Designing an effective network forensic framework for the investigation of botnets in the Internet of Things (Doctoral dissertation, UNSW Sydney).

[18] Roopak, M., Tian, G.Y. and Chambers, J., 2020, January. An intrusion detection system against ddos attacks in iot networks. IEEE, In 2020 10th annual computing and communication workshop and conference (CCWC) (pp. 0562-0567).

[19] Parra, G.D.L.T., Rad, P., Choo, K.K.R. and Beebe, N., 2020. Detecting Internet of Things attacks using distributed deep learning. Journal of Network and Computer Applications, 163, p.102662.

[20] Alsoufi, M.A., Razak, S., Siraj, M.M., Nafea, I., Ghaleb, F.A., Saeed, F. and Nasser, M., 2021. Anomaly-based intrusion detection systems in iot using deep learning: A systematic literature review. Applied sciences, 11(18), p.8383.

[21] Yue, C., Wang, L., Wang, D., Duo, R. and Yan, H., 2021. Detecting Temporal Attacks: An Intrusion Detection System for Train Communication Ethernet Based on Dynamic Temporal Convolutional Network. Security and Communication Networks, 2021, pp.1-21.

[22] Ge M., X. Fu, N. Syed, Z. Baig, G. Teo, and A. Robles-Kelly, "Deep learning-based intrusion detection for IoT networks," in Proc. 24th IEEE Pacific Rim Int. Symp. Dependable Comput. (PRDC), Dec. 2019, pp. 256–265.

[23] Pecori R., A. Tayebi, A. Vannucci, and L. Veltri, "IoT attack detection with deep learning analysis," in Proc. Int. Joint Conf. Neural Netw. (IJCNN), Jul. 2020, pp. 1–8.

[24] Idrissi I., M. Azizi, and O. Moussaoui, "IoT security with deep learning based intrusion detection systems: A systematic literature review," in Proc. 4th Int. Conf. Intell. Comput. Data Sci. (ICDS), Oct. 2020, pp. 1–10.

[25] D. Li, L. Deng, M. Lee, and H. Wang, "IoT data feature extraction and intrusion detection system for smart cities based on deep migration learning," Int. J. Inf. Manage., vol. 49, pp. 533–545, Oct. 2019.

[26] C. Yin, S. Zhang, J. Wang, and N. N. Xiong, "Anomaly detection based on convolutional recurrent autoencoder for IoT time series," IEEE Trans. Syst., Man, Cybern. Syst., early access, Feb. 7, 2020.

[27] Schuldt, C., Laptev, I. and Caputo, B., 2004, August. Recognizing human actions: a local SVM approach. In Proceedings of the 17th International Conference on Pattern Recognition, 2004. ICPR 2004. (Vol. 3, pp. 32-36). IEEE.

[28] Chauhan, V.K., Dahiya, K. and Sharma, A., 2019. Problem formulations and solvers in linear SVM: a review. Artificial Intelligence Review, 52(2), pp.803-855.

[29] Jakkula, V., 2006. Tutorial on support vector machine (svm). School of EECS, Washington State University, 37(2.5), p.3.

[30] Shahin, M., Chen, F.F., Bouzary, H., Hosseinzadeh, A. and Rashidifar, R., 2022. A novel fully convolutional neural network approach for detection and classification of attacks on industrial IoT devices in smart manufacturing systems. The International Journal of Advanced Manufacturing Technology, 123(5-6), pp.2017-2029.

[31] Leevy, J.L., Khoshgoftaar, T.M. and Peterson, J.M., 2021, August. Mitigating class imbalance for iot network intrusion detection: a survey. In 2021 IEEE Seventh International Conference on Big Data Computing Service and Applications (BigDataService) (pp. 143-148). IEEE.

[32] Banaamah, A.M. and Ahmad, I., 2022. Intrusion Detection in IoT Using Deep Learning. Sensors, 22(21), p.8417.

[33] Elejla, O.E., Belaton, B., Anbar, M., Alabsi, B. and Al-Ani, A.K., 2019. Comparison of classification algorithms on ICMPv6-based DDoS attacks detection. In Computational Science and Technology (pp. 347-357).

[34] Badotra, S. and Panda, S.N., 2021. SNORT based early DDoS detection system using Opendaylight and open networking operating system in software defined networking. Cluster Computing, 24, pp.501-513.

[35] Lilhore, U.K., Poongodi, M., Kaur, A., Simaiya, S., Algarni, A.D., Elmannai, H., Vijayakumar, V., Tunze, G.B. and Hamdi, M., 2022. Hybrid model for detection of cervical cancer using causal analysis and machine learning techniques. Computational and Mathematical Methods in Medicine, 2022.

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